Laboratory 3: Quantum Key Distribution

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ABSTRACT In this report we analyzed experimental data acquired during a laboratory session on Quantum Key Distribution (QKD), focusing on the estimation of relevant parameters for classical post-processing (QBER and SKR). The experimental setup is based on an existing and already implemented QKD protocols, in particular a 3-states, 1-decoy efficient BB84 protocol. States data are saved both for the sender and the receiver while sent decoy are stored on the sender. Additionally state absolute position is saved to evaluate the secrecy key rate in a finite key scenario in function of the time.

I. EXPERIMENTAL SETUP

To analyze the setup for this laboratory it could be better to study the two protocol subject separately.

A. SENDER (ALICE)

The sender (Alice) is controlled by a Field Programmable Gate Array (FPGA), which provides the system clock and the synchronization signals for state preparation. The FPGA operates with a clock period of $20\,\mathrm{ns}$, corresponding to a rate of $50\,\mathrm{MHz}$, and coordinates the emission of optical pulses as well as the modulation required for quantum state encoding and decoy-state preparation.

The optical source operates at a wavelength of $1550\,\mathrm{nm}$, chosen for compatibility with standard telecommunication fibers and low transmission loss. The laser emits short optical pulses that are initially prepared in a polarization state that is $\frac{1}{\sqrt{2}}(|H\rangle + |V\rangle)$. This preparation allows subsequent phase modulation to generate the required BB84 states.

Phase encoding is implemented using a phase modulator based on an iPOGNAC architecture. In this configuration, the optical pulse propagates in both clockwise and counter-clockwise directions within a fiber loop, applying phase shifts that depend on the modulation signal applied by the FPGA. The symmetry of the iPOGNAC geometry provides passive compensation of phase noise and polarization drifts, ensuring high stability. By applying appropriate phase shifts, Alice prepares the quantum states required for the efficient BB84 protocol. The output state is defined as $|\psi_{out}\rangle = \frac{1}{\sqrt{2}}(|H\rangle + e^{i(\phi_H - \phi_V)}\,|V\rangle)$ where the $\phi_H + \phi_V$ is the phase difference.

The iPOGNAC modulator is connected using polarization-maintaining (PM) fiber (to preserve the polarization alignment). Decoy states are implemented by varying the mean photon number of the emitted pulses. In this experiment, two decoy intensities are used, with mean photon numbers set to $\mu_1 := 0.6$ and $\mu_2 := 0.17$, respectively.

Following state preparation, the optical pulses pass through a Variable Optical Attenuator (VOA), which reduces the optical power to the single-photon level required for quantum key distribution and enables precise control of the decoy-state intensities. Finally, a broadband optical filter is placed at the output of the sender to limit the spectral bandwidth of the emitted pulses to that of the communication channel.

B. RECEIVER (BOB)

The receiver (Bob) is designed to perform polarization-resolved single-photon detection. The discrimination of the incoming quantum states is achieved through time-of-arrival measurements using a single Time-to-Digital Converter (TDC), which records the detection times of photons with respect to the system clock provided by the FPGA.

Since the standard single-mode optical fiber connecting Alice and Bob does not preserve polarization, the input of Bob, the optical signal is processed by an automatic polarization controller (APC), which is required to counteract slow drifts and random rotations induced by the fiber channel. Correct polarization alignment is essential, as all subsequent optical components assume a well-defined polarization basis.

After polarization compensation, the incoming signal is routed to the two measurement bases using a passive beam splitter. The receiver operates in an asymmetric configuration, where the majority of the times it measures using the basis used for key generation, while a smaller fraction it uses complementary basis employed for parameter estimation and security. With this basis choice maximizes the achievable secret key rate while preserving sufficient statistics to accurately estimate error rates and bound the information accessible to an eavesdropper.

The measurement in each basis is implemented through time-bin encoding. Instead of employing multiple single-photon detectors at the outputs of polarizing beam splitters, selected optical paths are delayed by well-defined temporal

offsets. These delays are chosen to be shorter than the system clock period of 20 ns, ensuring that all detection events corresponding to a given transmitted pulse occur within the same clock cycle. As a result, different measurement bases and logical outcomes are mapped onto distinct temporal detection windows.

All delayed optical paths are subsequently recombined and directed onto a single InGaAs single-photon avalanche diode (SPAD) operating at wavelength of 1550 nm. The detector is operated in gated mode and actively cooled to reduce dark counts and afterpulsing effects, which are particularly significant at this wavelength due to the small bandgap of the detector material. Although the use of a single detector introduces additional optical losses compared to multi-detector architectures, this design choice substantially reduces system complexity and mitigates cost.

The electrical output of the SPAD is connected to the TDC, which records the arrival time of each detection. Since Alice emits optical pulses every 20 ns, the arrival times can be associated with discrete time slots corresponding to individual transmitted pulses.

C. QUANTUM COMMUNICATION CHANNEL

The quantum communication channel connecting the sender and receiver consists of a standard single-mode optical fiber commonly used in classical telecommunication networks. As it reflects realistic deployment conditions for quantum key distribution systems.

II. DATA STORAGE & STRUCTURE

In our implementation of the efficient BB84, all relevant experimental data are stored in a compact format that allows full reconstruction of the transmitted and received quantum signals. The stored data encode the quantum states prepared by Alice, the associated decoy levels, the temporal positions of the transmitted pulses, and the states measured by Bob.

A. STATE/DECOY FILES

In the first file format, each byte defines a single quantum state or decoy level. Although only two bits are required to encode the information, the data are embedded into bytes for simplicity.

1) states file format

The quantum states prepared by Alice are stored in the `states.txt` while Bob's received states are stored in `statesRCV.txt`. Both file follows the same format:

H	'b00'	V	'b01'
D	'b10'	A	'b11'

Each state is embedded into an 8-bit byte, where the two least significant bits contain the state encoding. For example, the state D is stored as `00000010`.

2) decoys file format

The decoy intensity associated with each transmitted state is stored in the file `decoy.txt` and encoded as

null	'00'
low	'01'
high	'10'

In practice, no null decoy levels are expected in this file.

B. QBIT POSITION

The temporal position of the transmitted states are stored by Alice in `bitpos.txt`, the data are grouped into packets of 24,000 states. Each packet starts with the value -1, followed by the absolute position of the first state in the packet. For the remaining 23,999 states, only the relative difference with the previous state is stored. Each value is written on a separate row, so the file can be read as a standard one-column CSV file. Each unit in the bit-position file corresponds to a time interval of 20 ns, which reflects the system repetition rate of 50 MHz.

III. DATA PREPROCESSING

A. BEFORE DATA ANALYSIS

We assume that the raw experimental data has already been pre-processed at acquisition time. In particular, all time filtering, coincidence identification, and noise rejection procedures have been applied prior to the data being provided for analysis. Consequently, the data set contains only valid detection events.

B. WINDOW DEFINITION AND BLOCK SIZE

The data analysis is performed by dividing the full data set into consecutive blocks of fixed duration. Let T_{win} denote the window size of each block; throughout this work, we choose

$$T_{\text{win}} \in \{5\ \text{s}, 10\ \text{s}, 15\ \text{s}, 20\ \text{s}\} \tag{1}$$

Let $\Delta t = 20$ ns be the unit step size of the transmission. The size of the unit Δt in each block is defined as:

$$N_{\text{block}} = \frac{T_{\text{win}}}{\Delta t} \tag{2}$$

Using the bit-position file (`bitpos`), which records the emission index or time index of each prepared quantum state, we identify all state indices $i \in \{0, ..., k\}$ that fall within a given time window. In particular $t_k - t_0 \sim N_{\text{block}}$

C. STATE MATCHING AND EVENT SELECTION

For each state index belonging to a given block, we extract the corresponding information from the state file, including the prepared (sent) quantum state and the measurement outcome (received state). We then compare the preparation basis with the measurement basis.

Only events for which the sent and received bases coincide are retained for further processing, while all other events are discarded. This basis reconciliation step ensures that the remaining data is compatible with the parameter estimation procedures of the BB84 protocol.

At last using the information obtained from the decoy file we are able to differentiate the statistics for $\mu_1 = 0.6$ and $\mu_2 = 0.17$.

D. STATISTICAL DATA STRUCTURE

From the filtered events within each block, we compute a set of statistical quantities that characterize the observed data:

- $n_Z^{(\mathrm{s})}$ and $n_X^{(\mathrm{s})}$ as the number of total states sent by Alice in the Z and X basis;
- $n_{\mu,Z}^{(\mathrm{r})}$ and $n_{\mu,X}^{(\mathrm{r})}$ as the number of detection events in the Z and X basis for intensity level μ;
- $m_{\mu,Z}^{(\mathrm{err})}$ and $m_{\mu,X}^{(\mathrm{err})}$ as the number of bit errors observed in the Z and X basis for intensity level μ.

In addition, the corresponding basis preparation probabilities are then given by

$$p_Z = \frac{n_Z^{(\mathrm{s})}}{n_Z^{(\mathrm{s})} + n_X^{(\mathrm{s})}}, \qquad p_X = \frac{n_X^{(\mathrm{s})}}{n_Z^{(\mathrm{s})} + n_X^{(\mathrm{s})}}. \tag{3}$$

All quantities defined in this subsection are evaluated independently for each block and constitute the input data for the subsequent parameter estimation and secret key rate analysis.

IV. QBER ANALYSIS

V. QUANTUM BIT ERROR RATE (QBER)

The quantum bit error rate (QBER) is a central figure since it provides a quantitative measure of the noise present in the quantum channel and reflects the overall quality of the experimental setup. In addition, the QBER directly impacts the efficiency of the classical post-processing stage. In particular, a higher QBER implies a larger amount of information leakage during information reconciliation, since more parity information must be revealed to correct errors between the communicating parties. Consequently, an accurate estimation of the QBER is essential for both parameter estimation and secret key rate evaluation.

In this work, the QBER is evaluated independently for each block defined in Sec. III-B, and for each measurement basis used in the protocol.

A. QBER ESTIMATION IN Z,X

The QBER in basis b for a given block is defined as

$$\mathrm{QBER}_b = \frac{n_b^{(\mathrm{err})}}{n_b^{(\mathrm{r})}}. \tag{4}$$

The Z-basis QBER, QBER_Z, characterizes the error rate of the raw, therefore directly relevant for information reconciliation. The X-basis QBER, QBER_X, provides information about the phase error rate.

B. BLOCK ANALYSIS

For the experimental analysis of we chose a T_{win} of 5s.

VI. SECRET KEY RATE ANALYSIS

In this section, we describe the methodology used to estimate the secret key rate of the implemented three-state, one-decoy BB84 protocol. We first recall the general principles of secret key rate estimation in the asymptotic, infinite-key regime using the decoy-state method. We then discuss the modifications required to account for the finite-size nature of the experimental data and the absence of a vacuum decoy state.

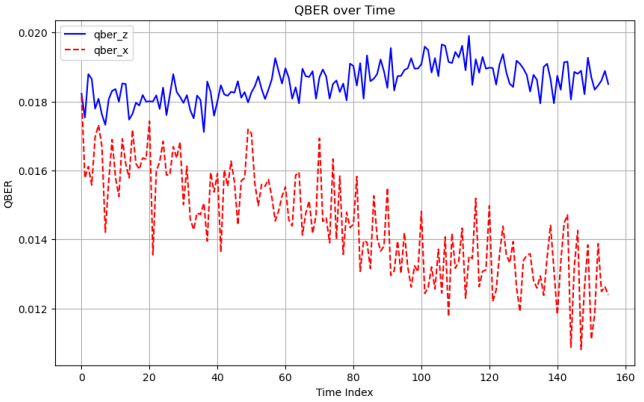


FIGURE 1. Qber analysis report for $T_{\mathrm{win}} = 5s$

A. SECRET KEY LENGTH FOR THE 3 STATE, 1-DECOY METHOD

It could be insightful to first look at a formulation that also allow the metrics of the null state to be calculated:

The key length can be written in the general form:

$$\begin{aligned} l \leq & s_{Z,0}^{l} + s_{Z,1}^{l}(1 - h(\phi_Z^u)) \\ & - \lambda_{EC} - a\log_2(b/\epsilon_{sec}) - \log_2(2/\epsilon_c or) \end{aligned} \tag{5}$$

where:

- $s_{Z,\mu}^{l}$ is the lower bound on events in which Bob had a detection and the pulse sent by Alice has intense μ
- ϕ_Z^u is the upper bound on the phase error rate (ϕ_Z)
- λ_{EC} is the number of bit disclosed for the error correction.
- $a, b = (6, 19)$ for this particular implementation

B. FINITE-SIZE EFFECTS AND EXPERIMENTAL CONSTRAINTS

The secret key must be extracted from a finite data set, divided into blocks of finite size, as described in Sec. III-B. As a consequence, the quantities appearing in the asymptotic key rate formula cannot be directly accessed and must be estimated from finite statistics.

Moreover, this implementation cemploys a single decoy intensity and does not include a vacuum decoy state. The absence of a vacuum state prevents the direct measurement of certain parameters and requires the use of analytical bounds to estimate the relevant single-photon quantities.

To address both the finite-size effects and the lack of direct access to some photon-number components, we replace the asymptotic quantities by conservative upper and lower bounds.

C. BOUNDS ON MEASUREMENT STATISTIC

The estimation of detection rates and error rates from finite data is performed using statistical inequalities. In particular, we invoke Hoefding inequality to relate the observed detection in each block to the corresponding asymptotic probabilities, up to a prescribed failure probability ε_1. In particular, we cand define $\delta(x, \varepsilon_1)$ as:

$$\delta(x, \varepsilon_1) := \sqrt{x \log(1/\varepsilon_1)/2} \tag{6}$$

In this way we can calculate the upper and lower bound on the detection and on the errors.:

$$\begin{aligned} &\forall k \in \{\mu_1, \mu_2, \mu_3\} \\ &\forall b \in \{Z, X\} \\ n_{b,k} =& \frac{e^k}{p_k}(n_{b,k}^{measure} \pm \delta(n_b, \varepsilon_1)) \\ m_{b,k} =& (m_{b,k} \pm \delta(m_b, \varepsilon_1)) \end{aligned} \tag{7}$$

D. PHOTON-NUMBER STATISTICS

The photon-number distribution of the emitted pulses is assumed to follow a Poisson distribution with mean photon number μ. The probability of emitting a pulse containing exactly k photons is given by

$$\tau_k = \Pr(k) = \frac{e^{-\mu}\mu^k}{k!}. \tag{8}$$

In particular, we define

$$\tau_0 = e^{-\mu}, \qquad \tau_1 = \mu e^{-\mu}, \tag{9}$$

as the probabilities of emitting a vacuum state and a single-photon state, respectively. These quantities enter explicitly into the decoy-state bounds and are known from the source calibration.

E. VACUUM AND SINGLE-PHOTON DETECTIONS

Using the finite-size corrected detection and error statistics, together with the photon-number probabilities defined above, we derive upper and lower bounds on the vacuum and single-photon contributions. These bounds allow us to estimate the minimum number of single-photon detection events that are compatible with the observed data.

The explicit expressions for the vacuum and single-photon bounds are introduced in the following subsection and are used to compute a lower bound on the achievable secret key rate in the finite-key regime.

$$s_{b,0} \leq s_{b,0}^u := 2(m_b + \delta(n_b, \varepsilon_1)) \tag{10}$$

The previous bound is a looser bound on the void event, leading to a lower final key lenght. By implementing this last result we are able to calculate $s_{b,0}^l$:

$$s_{b,0} \geq s_{b,0}^l := \frac{\tau_1\mu_1}{\mu_2(\mu_1 - \mu_2)}(n_{Z,\mu 2}^- - \frac{\mu_2^2}{\mu_1^2}n_{Z,\mu_1}^+ - \frac{(\mu_1^2 - \mu_2^2)}{\mu_1^2}\frac{s_{b,0}^u}{\tau_0}) \tag{11}$$

Therefore we are only missing the lower bound on $s_{Z,0}$ that is define as:

$$s_{Z,0} \geq s_{Z,0}^l := \frac{\tau_0}{\mu_1 - \mu_2}(\mu_1 n_{Z,\mu_2}^- - \mu_2 n_{Z,\mu_1}^+) \tag{12}$$

F. PHASE ERROR RATE

Similarly as before we will be using the above corrected detection to calculate the phase error ϕ_Z^u and we can do so by evaluating the phase error on X, since $\phi_Z \leq \phi_X$ and:

$$\phi_Z \leq \phi_X^u := \frac{v_{X,1}^u}{s_{X,1}^l} + \gamma(\varepsilon_s ec, \frac{v_{X,1}^u}{s_{X,1}^l}, s_{Z,1}^l, s_{X,1}^l) \tag{13}$$

where:

- $v_{X,1}^l := \frac{\tau_1}{\mu_1 - \mu_2}(m_{x,\mu_1}^+ - m_{X,\mu_2}^-)$
- ε_{sec} security parameter of the protocol

G. ERROR CORRECTION LEAKAGE

The amount of information revealed during error correction is upper bounded by

$$\lambda_{\mathrm{EC}} = f_{\mathrm{EC}}\, N_Z\, h(e_Z)\,, \tag{14}$$

where $N_Z = \sum n_Z$ denotes the total number of detection events in the Z basis, e_Z is the observed quantum bit error rate in the Z basis, and h(x) is the binary entropy function. The factor $f_{\mathrm{EC}} = 1.16$ accounts for the inefficiency of the practical error-correction algorithm relative to the Shannon limit. This expression provides a conservative upper bound on the information leaked to a potential eavesdropper during reconciliation.

VII. EXPERIMENTAL RESULT

A. KEY LENGTH PARAMETER

Parameter	Value
s_{z0}^u (vacuum upper bound)	2.083179×10^3
Upper bound on detections for n_z	2.890214×10^{-1}
τ_0 (vacuum probability)	5.812839×10^{-1}
τ_1 (single-photon probability)	3.090030×10^{-1}
s_{z0}^l (vacuum lower bound)	1.935358×10^4
s_{z1}^l (Z single lower bound)	2.622889×10^4
s_{x1}^l (X single lower bound)	1.948883×10^3
v_{x1}^u (phase error count)	3.825751×10^1
ϕ_z^u (phase error rate)	1.484394×10^{-2}
λ_{EC} (error correction)	6.242870×10^3
Security penalty	1.0×10^{-9}
Correctness penalty	1.0×10^{-15}
Final key length l	$\mathbf{3.616155 \times 10^4}$

TABLE 1. Verbose key length calculation summary for a chunck of 10s size

B. FINAL KEY LENGTH

In this subsection we are going to present the graph of the evaluated:

- Key length for a single test on the whole dataset with block length of 10s 2.
- The key length for different block size on a small number of blocks 3.

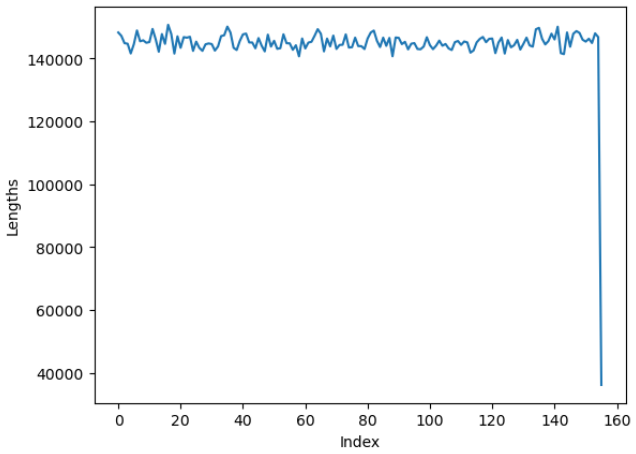


FIGURE 2. 157 blocks for a size of 10s per block

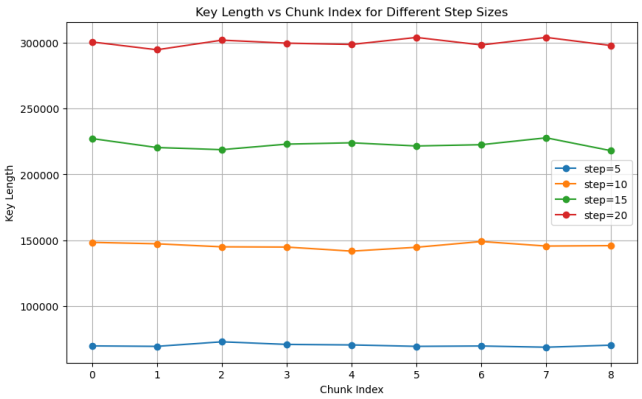


FIGURE 3. 15 blocks example of key length for different step size [5s, 10s, 15s, 20s]

C. FINAL KEY RATE

We calculated the final key rate by considering the total detection upper bound using the same Hoefding inequality presented in 7

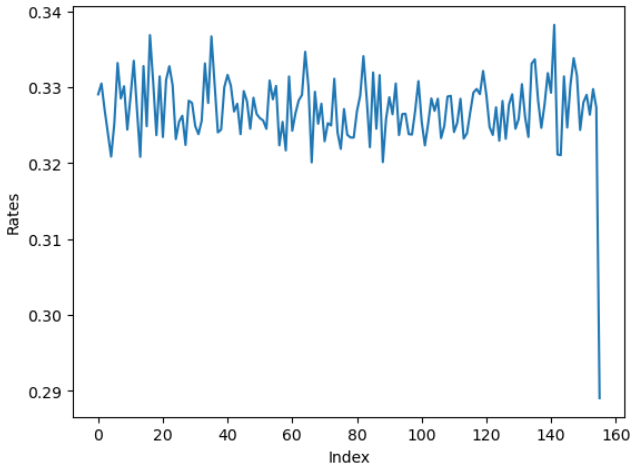


FIGURE 4. 157 blocks for a size of 10s per block

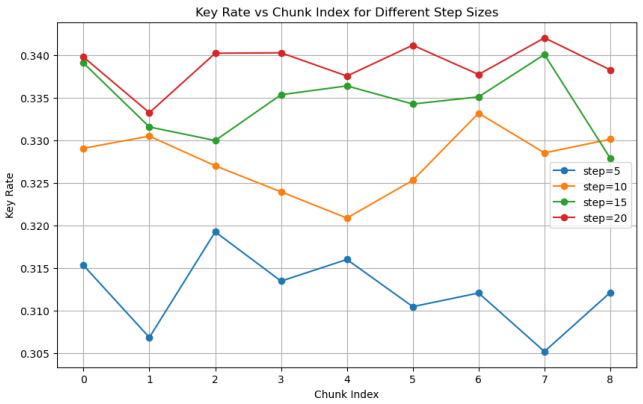


FIGURE 5. 15 blocks example of key rate for different step size [5s, 10s, 15s, 20s]

D. CONCLUSION

Analysis over 10s collected data The protocol operates in a low-noise regime, as evidenced by the small upper bound on the phase error rate, $\phi_Z^u = 1.48 \times 10^{-2}$, and a moderate error-correction leakage $\lambda_{\mathrm{EC}} \approx 6.2 \times 10^3$. Despite a large vacuum contribution ($\tau_0 \approx 0.58$), the single-photon component is tightly bounded, with $s_{z1}^l \approx 2.6 \times 10^4$, enabling efficient privacy amplification. As a result, the extracted secret key corresponds to approximately one half of the total detection events, demonstrating that high per-detection efficiency is achievable for a three-state, one-decoy protocol in the finite-key regime under favorable experimental conditions. Also the resulting secret key rate of approximately 0.34 bits per detection event is consistent with expectations for a three-state, one-decoy protocol operating in the finite-key regime, and confirms that the combined impact of phase-error estimation, error-correction leakage, and security overhead remains moderate under the present experimental conditions.

E. DECOY-STATE PROBABILITY MISMATCH AND PARAMETER SELECTION

In the implementation considered in this work, the decoy protocol specification provided by the instructor assumes fixed preparation probabilities of $p_{\mu_1} = 0.7$ for the signal intensity and $p_{\mu_2} = 0.3$ for the decoy intensity. However, an independent evaluation of the actual decoy statistics was performed using the log file generated by Alice (`decoy.txt`), which explicitly records the sequence of states sent during the experiment.

From this file, the empirically derived preparation probabilities are $p_{\mu_1}^{\mathrm{est}} \sim 0.88$ and $p_{\mu_2}^{\mathrm{est}} \sim 0.12$, revealing a significant discrepancy with respect to the nominal values. This difference has a substantial impact on the finite-key analysis:

- when the nominal probabilities $(0.7, 0.3)$ are used together with the measured experimental parameters, the resulting secret key length becomes negative, indicating an overly pessimistic parameter estimation.

- In contrast, employing the empirically estimated probabilities yields a positive and physically meaningful secret key length, consistent with the observed low-noise operating regime.

Since the decoy-state file is generated by Alice, who prepares and transmits the quantum states, the empirically derived probabilities are considered as the most faithful representation of the actual experimental configuration. Nevertheless, the nominal probabilities provided by the instructor are explicitly acknowledged.